

WEB BASED HOSTEL MANAGEMENT SYSTEM

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December 2020



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December 2020



**This dissertation is submitted in partial fulfilment of the requirement
of the Degree of Bachelor of Information Technology (external) of the
University of Colombo School of Computing**

DECLARATION

I certify that this dissertation does not incorporate, without acknowledgement, any material previously submitted for a Degree or Diploma in any University and to the best of my knowledge and belief, it does not contain any material previously published or written by another person or myself except where due reference is made in the text. I also hereby give consent for my dissertation, if accepted, to be made available for photocopying and for inter-library loans, and for the title and summary to be made available to outside organizations.



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ABSTRACT

Visakha Vidyalaya is one of the leading girls' schools in Colombo. Every year about 150 students are taken through the grade 5 scholarship examination. Among those students nearly 30 are entering to the hostel since their hometown is very far away or some other reasons.

The hostel got lots of responsibilities and duties to handle the inside process starting the very first day of a student registration to the hostel. It includes payment handling, all the records of the person who is taking out the student of the hostel and also taking back to the hostel, student's records, hostel details, student attendance.

To accomplish these important duties of the hostel the management using manual method in daily basis. Most of the time it's very difficult to handle these processes since they come up with mistakes of the records and so on. Therefore, the main objective of this project is to overcome these problems and find them to easy way of handling the processes.

This system has been designed using Rational Unified Process (RUP) methodology and developed using PHP, MySQL, Apache web server, HTML, Cascading Style Sheets, JQuery and JavaScript technologies. Through this newly introduced web based system, department would maintain their daily office works effectively, efficiently and accurately in future.

ACKNOWLEDGMENTS

I would like to take this opportunity to thank everyone who helped me in various way to complete my final year project successfully.

First of all, my gratitude goes to the University of Colombo School of computing, BIT coordinator and the Project Examination Board for giving me the offer to obtain the degree. Secondly my gratitude goes to the Lecter panel of the University of Colombo school of computing for guiding and helping us by giving learning materials for the past few years.

Also my special gratitude goes to the supervisor of my project Mr.M.P Sarath Wijesinghe for guiding and supporting me to get the best outcome of my work.

I would like to thank Mrs. W.S.D Weerasinghe the Vice Principle of Visakha Vidyakaya, the Teacher in charge of the Hostel and the Staff of the Hostel for helping me to get the all details and clearing my doubts regarding the necessary subjects of the hostel.

Finally, my sincere gratitude goes to my family, my friends and the teachers who gave me courage helped in various way to complete the project successfully.

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LIST OF ACRONYMS

AJAX -	Asynchronous JavaScript and XML
BIT -	Bachelor of Information Technology
CSS -	Cascading Style Sheet
DBMS -	Database Management System
ERD -	Entity Relationship Diagram
HTML -	Hyper Text Markup Language
IT -	Information Technology
JS -	JavaScript
MVC -	Model View Controller
OOA -	Object Oriented Analysis
OOD -	Object Oriented Design
PHP -	PHP: HyperText Preprocessor
RUP	Rational Unified Process
SDLC -	System Development Life Cycle
SQL -	Structured Query Language
UML -	Unified Modelling Language
URL -	Universal Resource Locator
XML -	Extensible Mark-up Language

CHAPTER 1 - INTRODUCTION

1.1 Introduction to the Hostel of Visakha Vidyalaya

The school began under the name of “Buddhist Girl’s Collage” In Turret Road, Colombo and then relocated in Vjira Road, Colombo and renamed it as Visakha Vidyalaya. It Is the only girls’ school identified amongst the first National Schools in the Island.

The Hostel of Visakha Vidyalaya got a special place in the school since the students who are unable to attend school from home specially ones who enters the school through grade five scholarship examination are taken to board in the hostel. All the facilities and secure is highly provided by the hostel. Student are allowed to any extracurricular activities while staying in the hostel and they’re not allowed to go outside from the school without permitted guardian provide by their parents.

Inside hostel all the day today activities are done by a schedule, students are not allowed do works beyond the schedule. Warden is the guardian of the students. In above there is a teacher in charge and a vice principal allocated to the hostel. Their duties are to handle the main processes and the warden is received instructions from them. Students are got relevant dates to meet visitors and to going out from the hostel only with a guardian. The hostel of Visakha Vidyalaya is known as highly secured and well manage place for a student to grow up. That’s why this hostel is getting bigger and provided with more facilities.

1.2 Motivation for the Project

- Hostel of Visakha Vidyalaya is the place where I grew up for 6 years, while been there it was found many difficulties in the management since there manual paper work were not accreted and because of that student and parents found many difficulties including myself. I understood Hostel is the better way for mw to implant my system in order to complete the BIT Degree
- BIT Final Individual Project is a key component of the Bachelor of Information Technology degree program of the University of Colombo. It is necessary to

complete it with successfully and to take this as an opportunity to enhance my knowledge and skills which I have gained throughout the degree program.

- The current system in this School hostel is done by manually starting from the student admission. Since the management and the parties who are involving with this manual system find many difficulties during the whole process. This online system is developing to overcome those human errors and to control with less paper work.
- The system will provide a better interaction among the communities who take part in different ways to the process and it will make easier to handle the current ongoing system with the new proposed one.
- This system will provide outsiders specially for the parents who are intend to enter their children to get to know about the hostel and the situation going through. I thoroughly believe this online system is worth developing for the hostel since it is actually helps to fulfill the task.

1.3 Objectives and scope of the proposed system

Student admission handling

The first day of a new student entering to the hostel, the system catches all the necessary data about the student including their family details, school related details, visitors who can visit and take them home, etc. Later on the student can view their details through the system and make relevant changes by the authority parties.

Hostel fees payment handling

This is one of the main process in the hostel and it has been having many issues during the past period of time. The system will provide a better way to overcome these difficulties since it's generating a report regarding the payment done by the student, therefore it won't be a matter in further in case of some student came up with a dough full situation about the payment.

Student's visitors handling

Another main subject handling in a hostel is the visitors are coming to meet a student and taking them outside or taking them home. This process is handling through a book

by writing down the in time and out time for the relevant date. The new system easily provides quick access to the relevant field and constraints missing the data.

To eliminate unnecessary paper work

Unnecessary paper work such as payments recording, visitors handling books, in and out details of a student, etc... Can be eliminated and separately saved in their user accounts.

Speed up of searching data

Since the current system is done by the manual work it is very slow and hard to get a necessary in case of emergency, but the online system finds it easy and quick to look for a past data of a particular student in above mentioned documents and authorized access can have printouts of the certain documents. The system increase work efficiency and also can reduce errors.

To add images of the events and occasion in the hostel

Annual Hostel day and year end Get together parties are the main occasions celebrating by the hostellers. The gallery in this system provides to upload the pictures taken in the particular days.

To search for the contact details

When someone wants to contact or get basic details about the hostel is very easy to find it since the system is contained with all the necessary information related to hostel from top to bottom.

Generate reports and graphs

Admin can login and get report about payments, visitors, student in and out details and attendance (daily/weekly/monthly/yearly).

1.4 The Dissertation Structure

1.4.1 Chapter 02: Analysis

Analysis phase is the initial phase of the system development life cycle. The chapter is contained existing systems that are similar to the proposed system, the behavior of the

current system and the functional and non-functional requirements as well the chapter is included the methods used to gather the requirements.

1.4.2 Chapter 03: Design of solution

This chapter is contained the design strategies used to the overall system and the structure of the system as well as the different kind of diagrams and models used to describe the system in a better way.

1.4.3 Chapter 04: Implementation

This chapter is consisted of the implementation of the Hostel management system for the school. Database construction, coding of the modules and user interfaces are included in this topic. The chapter is accented with the server side and client side development to achieve the client expectations.

1.4.4 Chapter 05: Evaluation

This chapter is mainly focused on clarifying the test results. The system should follow an accordant test plan to prove that it has met the specifications.

1.4.6 Chapter 06: Conclusion

This is the last chapter and it is represented the achievements of the project and the future extensions to be included.

1.4.7 References

The information of the url which were used to complete the report are included under this topic.

1.4.8 Appendixes

Under the appendixes there are attached the further information of the above dissertation chapters which were not added.

CHAPTER 2 - SYSTEM ANALYSIS

2.1 Chapter Overview

The main objective of this chapter is to discuss about necessary information that should gathered before design the proposed system. Further analysis the existing system and identify problems and problem origins to provide solutions through the proposed system.

2.2 Current system Study

2.2.1 Introduction

The current system of Visakha Vidyalaya Hostel is done by manually. Starting from the registration process all the records related o hostel is conducting manually and store in papers. Since the records are not in a proper way to maintain it finds very difficult to search for an old data and take vast amount of time to reach for the relevant field. Therefore, web based hostel management helps to overcome these problems since all the records are store and handling a proper way.

Drawbacks of the existing system

- Currently student registration is done by manually and it is very hard get the full details about the students in the hostel in an immediate situation, also the staff found difficulties for searching an old data.
- In the method of students' monthly payment handling the student is received a bill and if sometimes the bill was misplaced student cannot prove the payment which was done.
- When a visitor comes to meet a student the security if the gate has to go for the all data relevant to that student to find which guardian has arrived and it takes more time.
- The hostel is having annul events such as hostel day and get together parties, but there is no any proper way to publish the images or event details.

2.2.2 Use Case Diagram for current system

The current system was examined by the use case diagram. Actors are the stakeholders who are connected with the system. Use cases are the functions the done by the current system

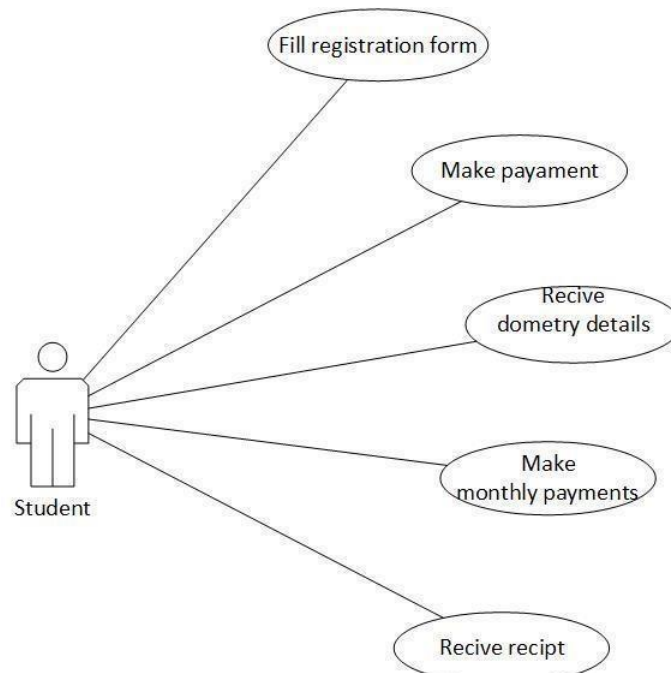


Figure 2. 1- Student Module

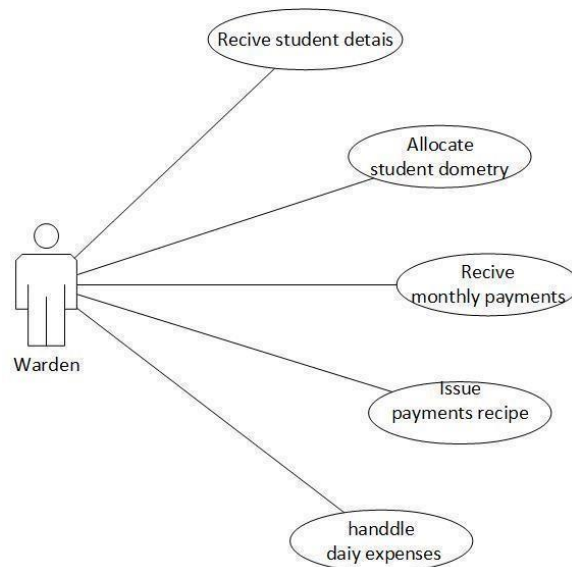


Figure 2. 2-Warden Module

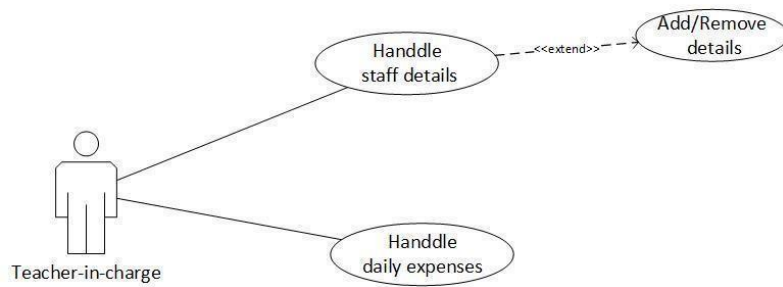


Figure 2. 3-Teacher-in-charge Module

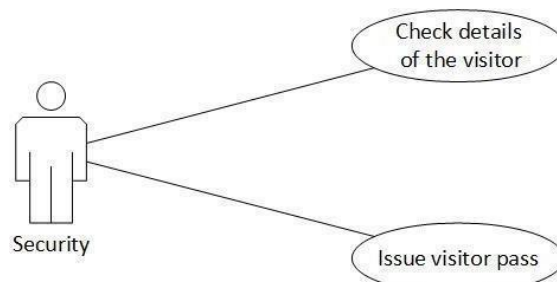


Figure 2. 4-Security Module

2.3 Fact Finding Techniques

Fact finding techniques are the methods that can use to gather information of company background & existing process. These are some of the techniques used in this project.

- Interviews
- Questionnaires
- Inspecting existing documents

2.3.1 Interviews

The knowledge about the whole process of the hostel is very important and it helps to find out the major processes which should be changed during the implementation. For this purpose, I had interviews with two of the high level management and the interview summaries are given below.

Interview

Interviewer	K.P Shayanika Sandaruwani
Designation	Vice principal of Visakha Vidiyalaya
Interviewee	W.S.D Weerasinghe
Venue	Visakha Vidiyalaya, 133, Vajira road, Colombo 05
Date	27 th of February 2020
Time Duration	1 Hour
Information & Decisions	• Current manual system. • Issues of the current system and the area expecting to change • Main processors which should be improved.

Table 2. 1 - Interview 1 Details

2.3.2 Questionnaires

Questionnaires are one of the popular fact finding technique that consisting of a series of questions can use to gather information from large number of people. It is the main advantage of the questionnaires. Different types of questionnaires were given to every grade and o the hostel staff to gather information about the existing system and he issues they face to.

2.3.4 Inspecting Existing Documents

Inspecting existing documents is a common fact finding technique that can use to understand the existing system easily. Some of the existing documents were inspected

before design the new system such as student registration, in and out documents, daily expenses.

2.4 Functional Requirements of the System

Functional requirements should include functions performed by specific screens, outlines of work-flows performed by the system, below are the functional requirements of the system recognized by the management level staff of the Hostel.

- The student could be registered to the hostel through online
- Parents could do the monthly payment through online
- The guardian details only can enter update delete by the higher level of management.
- The daily expenses of the hostel can be taken as a report by the system
- The system should allow to send notifications regarding the payments
- Inventory of the hostel can be seen any time to the staff as well
- System should provide facility to acknowledge about the events going in the hostel

2.5 Non-functional Requirements of the System

Non-functional requirements cover all the remaining requirements which are not covered by the functional requirements. They specify criteria that judge the operation of a system, rather than specific behaviors.[3]

These are the non-functional requirements included in this system.

- Security
- Efficiency
- Usability

2.5.1 Security

Monthly payment handling is a critical functionality in the hotel and after a payment is done it cannot be changed or removed without the permission of higher level management. Other authorized parties only can view it and search.

2.5.2 Efficiency

Searching for an old data is wasting time specially when student check-in and check-out details are very hard to find in the document. This system helps to find out easily for any data regarding to the hostel.

2.5.3 Usability

By providing very attractive user friendly interfaces system users can easily understand and have access to the system without any difficulties. This purpose is done by using fascinating UI elements and easily understandable components.

2.6 Existing Similar Systems

2.6.1 IFNOSS



Figure 2. 5-INFOSS logo and home page

Hostel management system offers a full featured accommodation system to efficiently manage the entire residential facility in the University of College. Reducing the staff& paper works, this accommodation software will keep the updated records of students, their meal, lodging, transfer room and other facilities. The software allocates rooms to the registered students based on their preferences and records their details of using food, lodging and other facilities along with fee and other advance payment details. The system also gathers a feedback on the room, food and other amenities on a regular basis and sends fee reminders to all registered students. This hostel management system monitors attendance, disciplinary logs, room charge status and check-in check-out in hostel premises.[1]

2.6.2 University of Kashmir

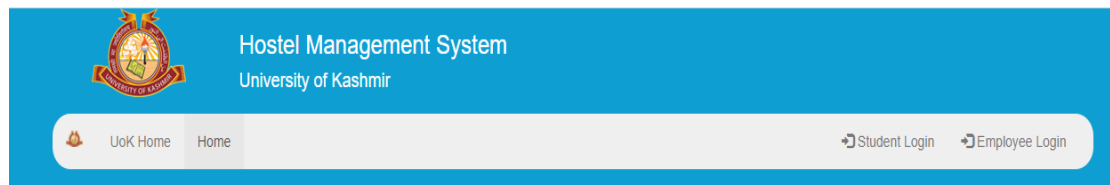


Figure 2. 6-University of Kashmir logo and home page

CHAPTER 3 - DESIGN

3.1 Chapter Overview

This chapter contains the Use Cases Diagram and relevant description of the proposed system, also the Class diagram and the Database structure using logical and conceptual data design techniques will be included.

3.2 Process Model

The goal for this chapter is to present the background and specific analysis techniques needed to construct a statistical model that describes a particular scientific or engineering process. The types of models discussed in this chapter are limited to those based on an explicit mathematical function. These types of models can be used for prediction of process outputs, for calibration, or for process optimization.[2]

3.3 Prototyping Model

Software prototyping is becoming very popular as a software development model, as it enables to understand customer requirements at an early stage of development. It helps get valuable feedback from the customer and helps software designers and developers understand about what exactly is expected from the product under development.[3]

3.4 Waterfall Model

The Waterfall Model was the first Process Model to be introduced. It is also referred to as a linear-sequential life cycle model. It is very simple to understand and use. In a waterfall model, each phase must be completed before the next phase can begin and there is no overlapping in the phases.[4]

3.5 Component Based Software Engineering (CBSE)

Component-based development (CBD) is a procedure that accentuates the design and development of computer-based systems with the help of reusable software

components. With CBD, the focus shifts from software programming to software system composing.

Component-based development techniques involve procedures for developing software systems by choosing ideal off-the-shelf components and then assembling them using a well-defined software architecture. With the systematic reuse of coarse-grained components, CBD intends to deliver better quality and output.[5]

3.6 Methodology for the proposed system

3.6.1 Rational Unified Process (RUP) Methodology

1. RUP is a software development process from Rational, a division of IBM. It divides the development process into four distinct phases that each involve business modeling, analysis and design, implementation, testing, and deployment. The four phases are:
2. **Inception** - The idea for the project is stated. The development team determines if the project is worth pursuing and what resources will be needed.
3. **Elaboration** - The project's architecture and required resources are further evaluated. Developers consider possible applications of the software and costs associated with the development.
4. **Construction** - The project is developed and completed. The software is designed, written, and tested.
5. **Transition** - The software is released to the public. Final adjustments or updates are made based on feedback from end users.[6]

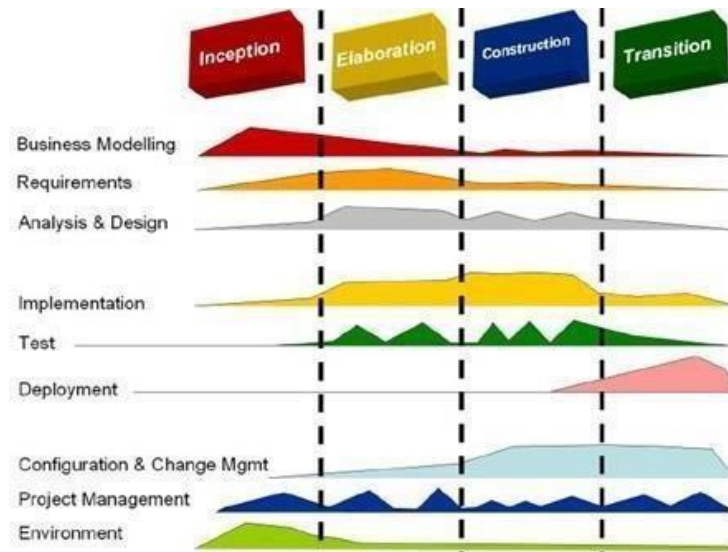


Figure 3. 1-Rational Unified Process [4]

3.7 Alternative solutions to the system

There are several options for clients who are expecting to have alternative solutions.

3.7.1 Free software packages

Using free software package may cause many problems to the client. The prototype which is published as a free version can contain bugs and the customers find less support from the developers.

3.7.2 Commercial standalone packages

This packages contains much better customer support in software. But in long run the customers may find problems since the enhancement of technology and platform independency. Customizations are less in these packages.

3.8 System as a web solution

The proposed system is developed as a web based solution instead of standalone application. Below are the key components can be find in the system.

- The implementation can be built easily.
- Without wasting time on installing one single deployment can be done on each

user computer.

- Remote access to the users.
- Any user can use it easily since the platform independence.

3.9 Object oriented analysis and designing in the system

Developing great software is never an easy process, especially if you are trying to figure out the best place to start. When developing software, you need to be able to understand what your end product (that's your software) is actually supposed to do and what sequence of steps you need to take in order to achieve this end goal.

This is where OOAD comes in. OOAD is often used in the area of object oriented programming (OOP). OOP aims to produce software that is efficiently written with few instances of duplicated code (not to mention code that is flexible and robust). The OOP approach also helps develop code that we can easily relate to the physical world we know and live in.[7]

3.10 Proposed System Use Case Diagram

A use case diagram is a graphic depiction of the interactions among the elements of a system. A use case is a methodology used in system analysis to identify, clarify, and organize system requirements. The actors, usually individuals involved with the system defined according to their roles. The use cases, which the specific roles are played by the actors within and around the system.[8]

3.10.1 Student Module

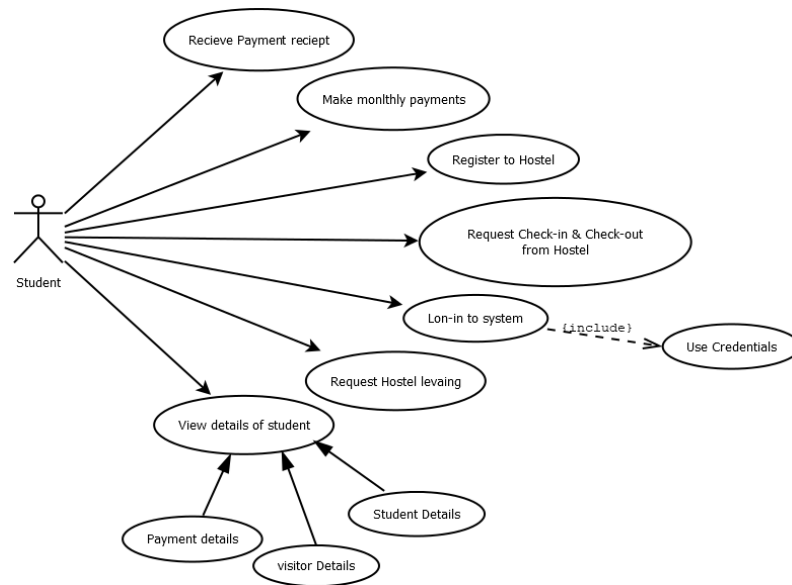


Figure 3. 2-Use Case Diagram for proposed System Student Module

3.10.2 Warden Module



Figure 3. 3-Use Case Diagram for proposed System Warden Module

3.10.3 Security Module

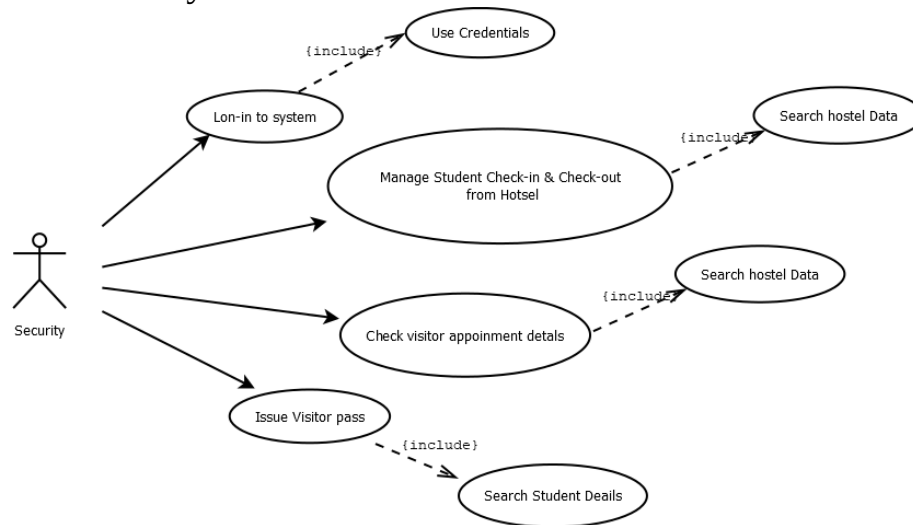


Figure 3. 4-Use Case Diagram for proposed System Security Module

3.10.4 Parent Module

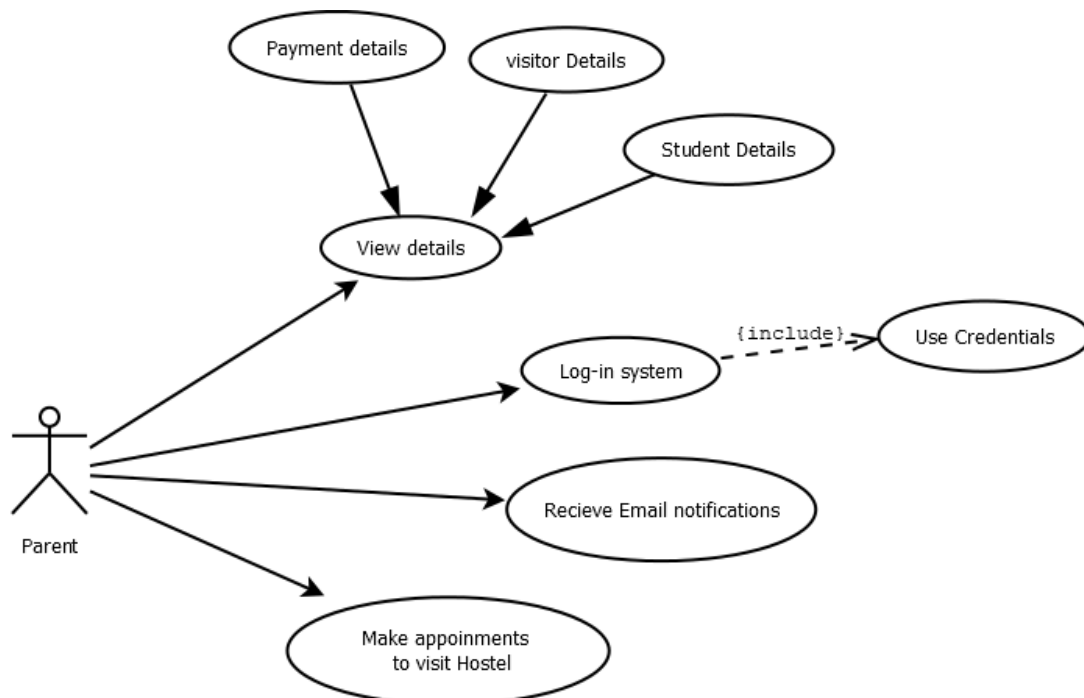


Figure 3. 5-Use Case Diagram for proposed System Parent Module

3.10.5 Visitor Module

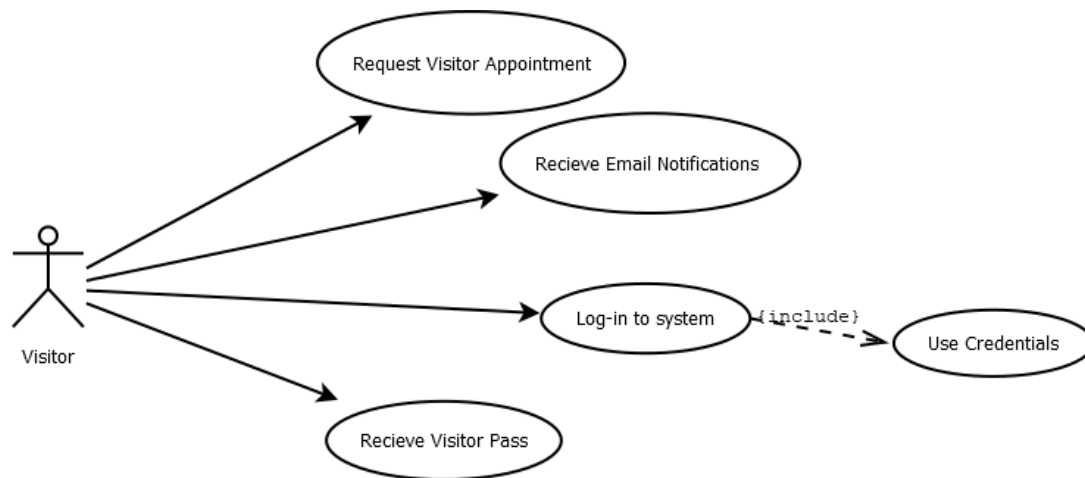


Figure 3. 6-Use Case Diagram for proposed System Visitor Module

3.10.6 Vice Principle Module

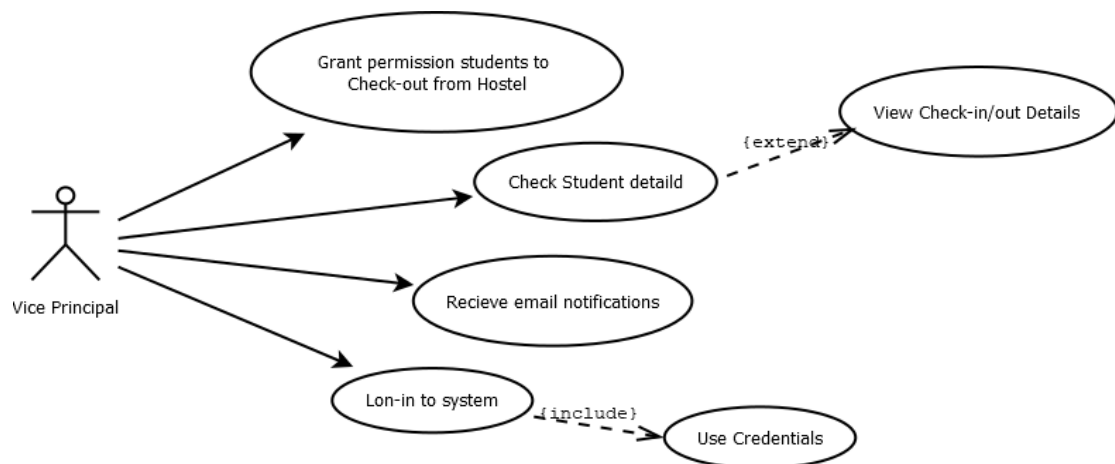


Figure 3. 7-Use Case Diagram for proposed System Vice Principle Module

3.11 Sequence diagram for the proposed system

Sequence diagrams describe interactions among classes in terms of an exchange of messages over time. They're also called event diagrams. A sequence diagram is a good way to visualize and validate various runtime scenarios. These can help to predict how a system will behave and to discover responsibilities a class may need to have in the process of modeling a new system.[9]

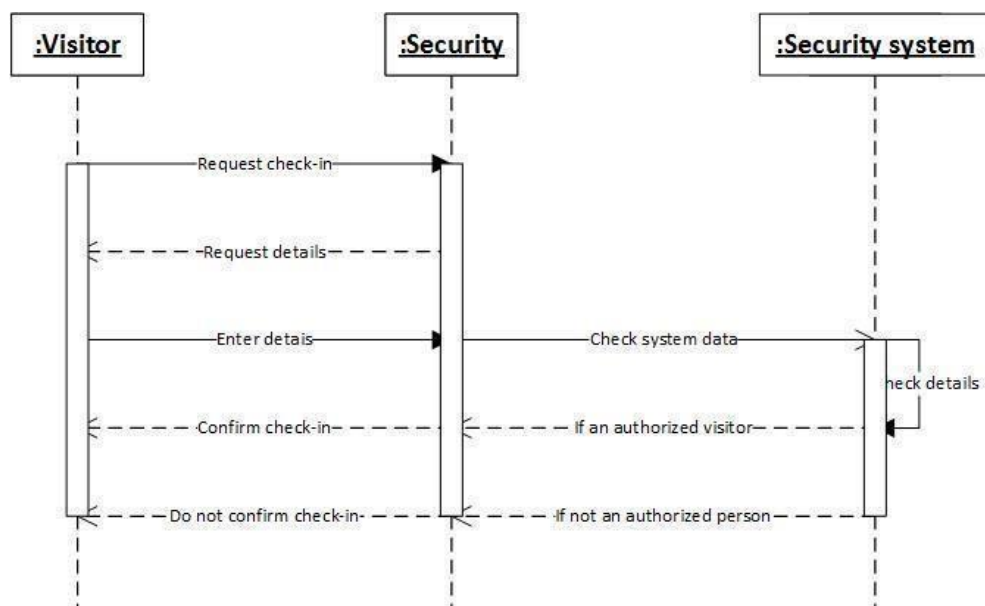


Figure 3. 8-Sequence Diagram for the Visitor check-in

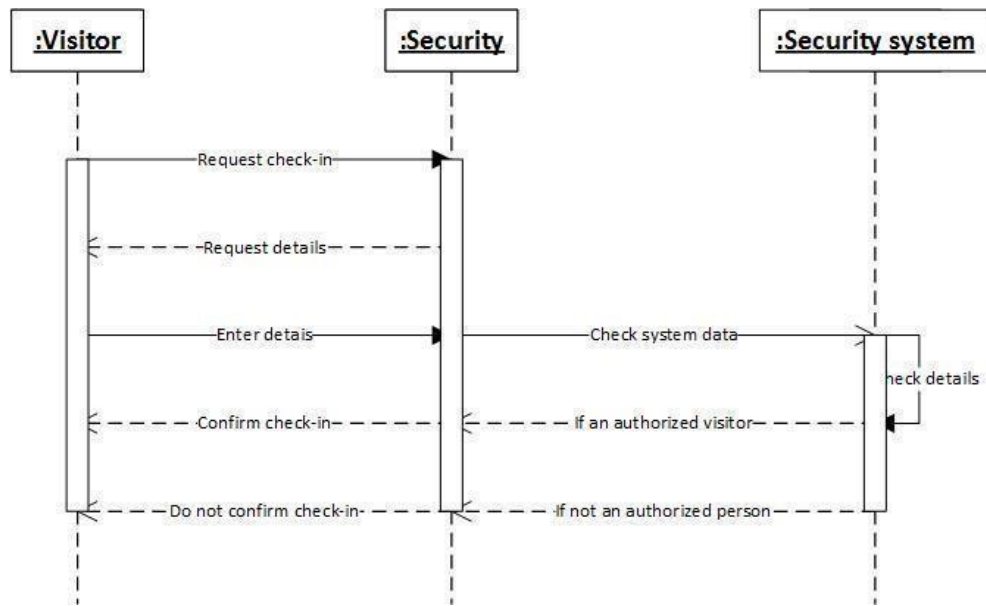


Figure 3. 9-Sequence Diagram for the Visitor check-in

3.12 Activity diagram for the proposed system

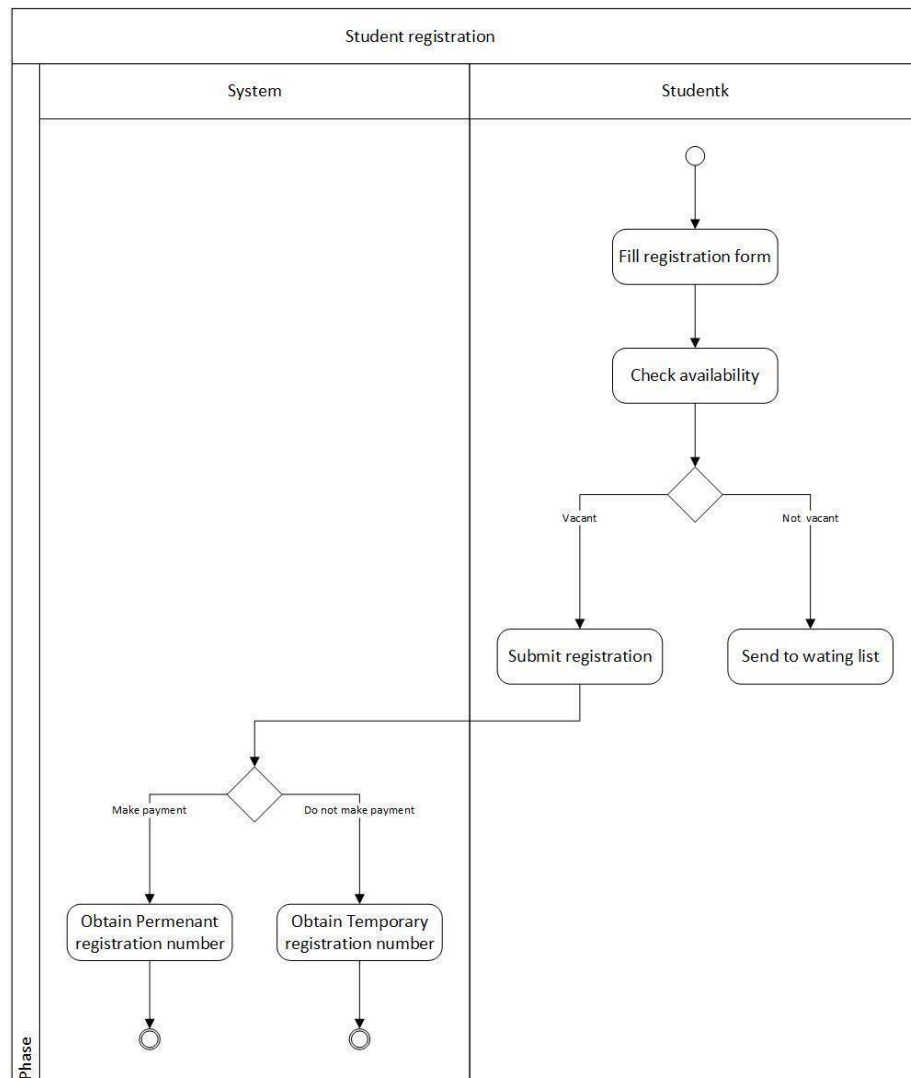


Figure 3. 10-Activity Diagram for the Student registration

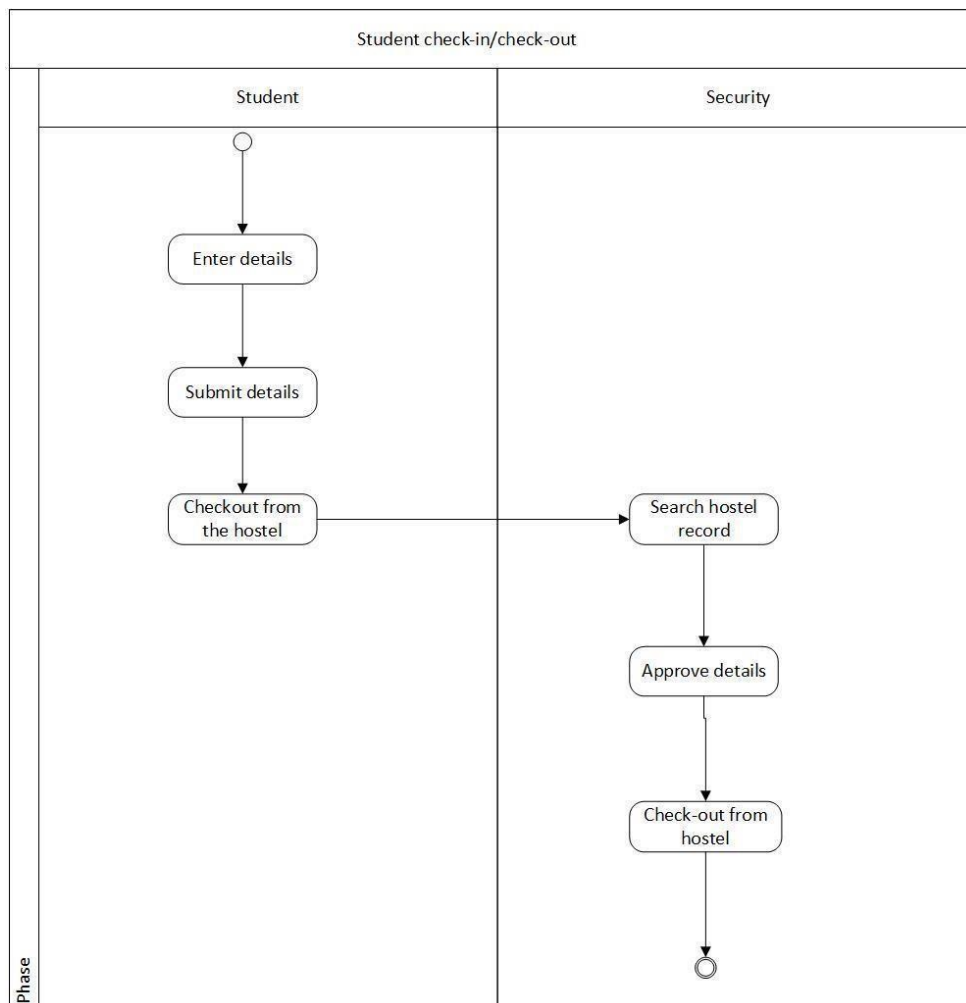


Figure 3. 11-Activity Diagram for the Student check-in/check-out

3.13 Entity Relationship Diagram for the proposed system

An entity relationship diagram (ERD) shows the relationships of entity sets stored in a database. An entity in this context is a component of data. In other words, ER diagrams illustrate the logical structure of databases.[8]

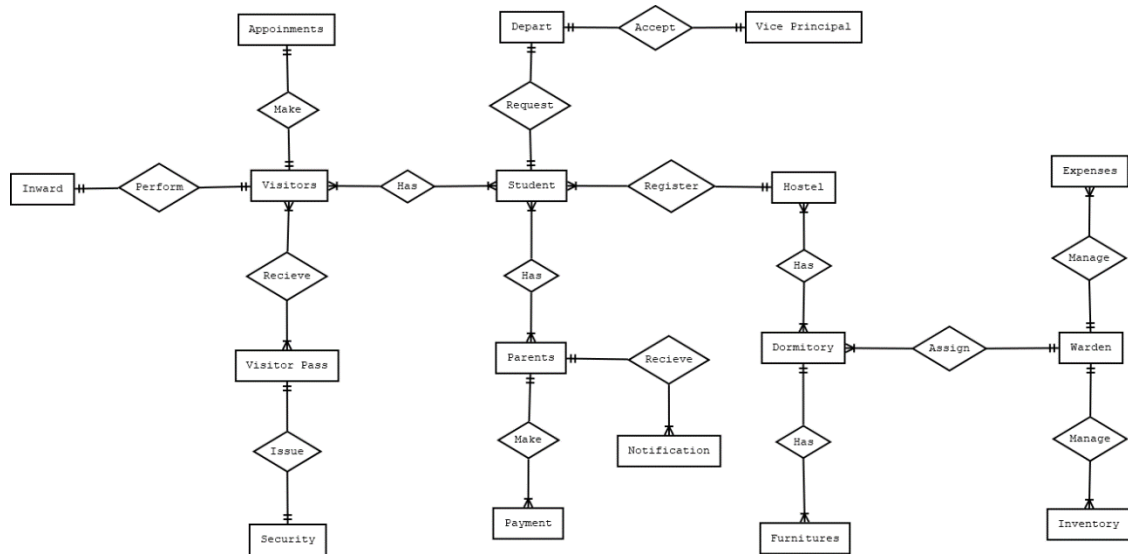


Figure 3. 12-Entity Relationship Diagram for proposed system

CHAPTER 4 - IMPLEMENTATION

4.1 Chapter Overview

In software development life cycle implementation stage plays a major role. The chapter is included with the tools and techniques used to develop the system and the reusable components used. Also the chapter is described about the facts which are necessary to implement the system.

4.2 Software Environment

4.2.1 Development Languages

- PHP (Hypertext Preprocessor)
- JavaScript
- JQuery
- HTML (HyperText Markup Language)
- CSS (Cascading Style Sheet)
- AJAX (Asynchronous JavaScript and XML)

PHP: Hypertext Preprocessor (or simply PHP) is a server-side scripting language designed for Web development, but also used as a general-purpose programming language. It was originally created by Rasmus Lerdorf in 1994, the PHP reference implementation is now produced by The PHP Group.[6] PHP originally stood for Personal Home Page, but it now stands for the recursive initialism PHP: Hypertext Preprocessor.

PHP code may be embedded into HTML code, or it can be used in combination with various web template systems, web content management systems, and web frameworks. PHP code is usually processed by a PHP interpreter implemented as a module in the web server or as a Common Gateway Interface (CGI) executable. The web server combines the results of the interpreted and executed PHP code, which may be any type of data, including images, with the generated web page. PHP code may also be executed with a

command-line interface (CLI) and can be used to implement standalone graphical applications.[11]

Hypertext Markup Language (HTML) is the standard markup language for creating web pages and web applications. With Cascading Style Sheets (CSS) and JavaScript, it forms a triad of cornerstone technologies for the World Wide Web.

Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document.

HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML provides a means to create structured documents by denoting structural semantics for text such as headings, paragraphs, lists, links, quotes and other items. HTML elements are delineated by tags, written using angle brackets. Tags such as `<img />` and `<input />` directly introduce content into the page. Other tags such as `<p>` surround and provide information about document text and may include other tags as sub-elements. Browsers do not display the HTML tags, but use them to interpret the content of the page.[12]

4.2.2 Development Frameworks

- Bootstrap

Bootstrap is a free and open-source front-end framework for designing websites and web applications. It contains HTML- and CSS-based design templates for typography, forms, buttons, navigation and other interface components, as well as optional JavaScript extensions. Unlike many earlier web frameworks, it concerns itself with front-end development only. Bootstrap is the second most-starred project on GitHub, with more than 126,000 stars.[13]

4.2.3 Other Supported Software, Development Tools and Browsers

- Windows 10 Operating System WAMP Server 2.4
 - Apache : 2.4.4

- MySQL : 5.6.12
- PHP My Admin : 4.0.4
- Netbeans IDE 8.2
- Web Browsers
 - Chrome
 - Firefox
 - Opera
 - and IE 8 above
- Design tools:
 - Adobe Photoshop CC
 - Adobe Illustrator
- For documentation:
 - Microsoft Word 2016
 - Microsoft Project 2013
 - Microsoft Visio 2013
 - Adobe reader X

WAMP is sometimes used as an abbreviated name for the software stack Windows, Apache, MySQL, PHP. It is derived from LAMP which stands for Linux, Apache, MySQL, and PHP. As the name implies, while LAMP is used on Linux servers, WAMP is used on Windows servers. Because WordPress isn't usually installed on Windows Servers, WAMP has become popular among developers as a method of installing WordPress on their personal computers.[14]

phpMyAdmin is a free and open source administration tool for MySQL and MariaDB. As a portable web application written primarily in PHP, it has become one of the most popular MySQL administration tools, especially for web hosting services.[15]

4.3 Hardware Specification

- Processor: 2.4 GHz Intel Core i3

- RAM: 4GB
- Hard Disk Capacity: 500GB

4.3.1 Architectural Overview

Like everything else in software engineering, it seems, the concept of Model- View- Controller was originally invented by Smalltalk programmers.

More specifically, it was invented by one Smalltalk programmer, Trygve Reenskaug. Trygve maintains a page that explains the history of MVC in his own words. He arrives at these definitions in a paper he published on December 10th, 1979:

Models

Models represent knowledge. A model could be a single object (rather uninteresting), or it could be some structure of objects.

There should be a one-to-one correspondence between the model and its parts on the one hand, and the represented world as perceived by the owner of the model on the other hand.

Views

A view is a (visual) representation of its model. It would ordinarily highlight certain attributes of the model and suppress others. It is thus acting as a presentation filter.

A view is attached to its model (or model part) and gets the data necessary for the presentation from the model by asking questions. It may also update the model by sending appropriate messages. All these questions and messages have to be in the terminology of the model, the view will therefore have to know the semantics of the attributes of the model it represents.

Controllers

controller is the link between a user and the system. It provides the user with input by arranging for relevant views to present themselves in appropriate places on the screen. It provides means for user output by presenting the user with menus or other means of

giving commands and data. The controller receives such user output, translates it into the appropriate messages and pass these messages on to one or more of the views.

4.3.2 Reused Modules

The following pre-coded modules were used while developing the system.

- PHPMailer - PHPMailer library was used to send emails

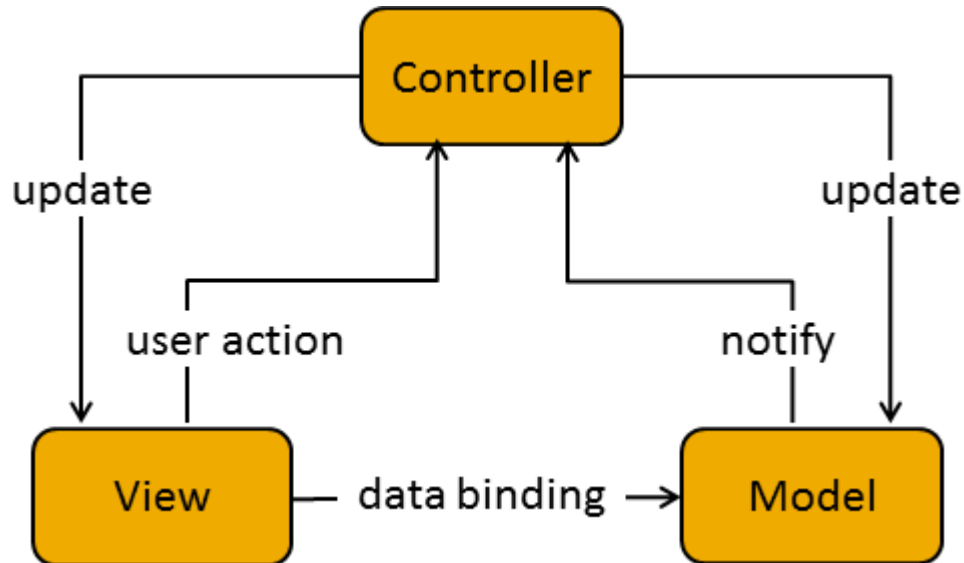


Figure 4. 1-MVC design diagram

- mPDF – mPDF library was used to generate relevant reports of the system.
- Bs-Admin-Bcore – The Bootstrap template was used to design the system.
- Sweet Alert - When building JavaScript-driven systems the feedback is provided the user to let them know if the action they have performed has been successful or not. In the early days, the window.alert() function is used to create messages by the web developers. While alert() works in practice and it's also consistent across browsers, it's not very flexible and its look and feel is not attractive to the user. function that shows very pretty modal windows. It's a standalone library that has no dependencies, and it's made from a JavaScript file plus a CSS file.
- JQuery Datepicker and FullCalendar 3.1.0
- JQuery Timepicker
- elfinder file manager

4.4 Code Structure

The following, figure represent the code structure.

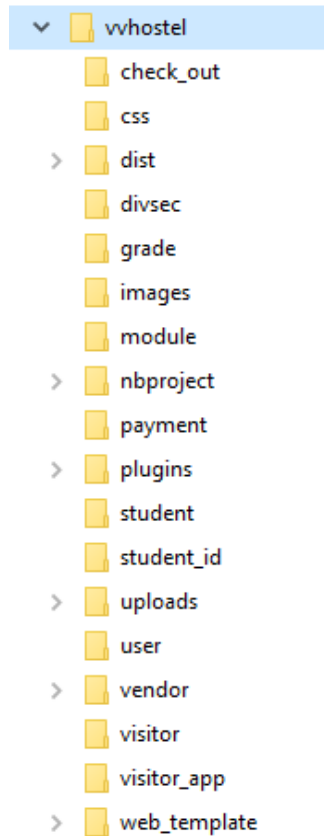


Figure 4. 2-Code Structure

4.4.1 Major Code Segments

Following are some of major code segments and some of them are code snippets since not showing full codes because of the space limit.

Database connection

```
<?php
$servername="localhost";
$username="root";
$password="";
$dbname="db_vvhostel";$conn = new mysqli($servername, $username, $password, $dbname);
if($conn->connect_error){
die("Connection Failed : ".$conn->connct_error);}?>
```

Code Snipped for log-in user

```
<?php
session_start();
include 'helper.php';
include 'conn.php';
if ($_SERVER["REQUEST_METHOD"] == 'POST') {
    $user_name = $password = null;
    $e = array();
    $user_name = clean_input($_POST['user_name']);
    $password = clean_input($_POST['password']);
    if (empty($user_name)) {
        $e['user_name'] = "The User name should not be empty..!";
    }
    if (empty($password)) {
        $e['password'] = "The Passord should not be empty..!";
    }
    if (empty($e)) {
        $sql = "SELECT * FROM tb_user WHERE user_name='$user_name' AND password='".
sha1($password) . "'";
        $result = $conn->query($sql);
```

```

if ($result->num_rows == 1) {
    while ($row = $result->fetch_assoc()) {
        $_SESSION['user_id'] = $row['user_id'];
        $_SESSION['first_name'] = $row['first_name'];
        $_SESSION['last_name'] = $row['last_name'];
        $_SESSION['profile_image'] = $row['profile_image'];
        $_SESSION['login_key'] = 1992;
    }
}

```

Insert data into Table

```

if (empty($e)) {
    $sql = "INSERT INTO tb_call_reg(`student_id`,`reg_date`,`reg_time`,
`email`,`reg_status`) VALUES "
        . "(" . $student_id . "," . $reg_date . "," . $reg_time . "," . $email . "," . 0) ";
    $r = $conn->query($sql);
    if ($r === TRUE) {
        foreach ($docs as $docs_code) {
            $sql = "INSERT INTO `tb_app_docs`(`student_id`,`docs_code`) VALUES
('$student_id','$docs_code')";
            $conn->query($sql);
        }
        foreach ($faci as $faci_code) {
            $sql = "INSERT INTO tb_app_faci(`student_id`,`faci_code`) VALUES
('$student_id','$faci_code')";
            $r = $conn->query($sql);
        }
        echo "Data saved..";
    } else {
        echo "Error..!" . $conn->error;
    }
}

```

Delete Data

```
<?php
if ($_SERVER["REQUEST_METHOD"] == 'POST' && isset($_POST['operate'])) {
    $_operate = $student_id = null;
    $operate = $_POST['operate'];
    $student_id = $_POST['student_id'];
    if ($operate == "yes") {
        $sql = "DELETE FROM tb_student WHERE student_id='$student_id'";
        $r = $conn->query($sql);
        if ($r === TRUE) {
            header("Location:view_call_reg.php");
        } else {
            echo "Error..! " . $conn->error;
        }
    } else {
        header("Location:view_call_reg.php");
    }
}
```

Java Script code snippet for load Divisional Secretariat

```
<script src="web_template/jquery/jquery.min.js"></script>
<script src="web_template/bootstrap/js/bootstrap.bundle.min.js"></script>
<script>

    function loadDivSec() {
        var mydata = $("#register").serialize();
        $.ajax({
            type: 'POST',
            data: mydata,
            url: 'divsec/divseccombo.php',
            success: function (data) {
```

4.5 Reused modules

Some of the well, tested frameworks and open source libraries were used while developing the system

- **Bootstrap v3.0** - The most popular and open source CSS framework in web development.
- **SMTP Mailer**- This is library which was used to send emails
- **M-Pdf**- mPDF library was used to report generation in PDF formats.
- **Date/Time picker** - This is one of JavaScript open source library which can handle events.
- **ChartJS** - This is also a popular JavaScript library for graphs.

CHAPTER 5 - EVALUATION

5.1 CHAPTER OVERVIEW

This chapter is consisted about the developed system's evaluation procedure. Test plan, test cases and user acceptance test results were carried out in order to ensure that all the aspects of the system have been tested.

5.2 SYSTEM TESTING

System testing of software or hardware is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black-box testing, and as such, should require no knowledge of the inner design of the code or logic.

As a rule, system testing takes, as its input, all of the "integrated" software components that have passed integration testing and also the software system itself integrated with any applicable hardware system(s). The purpose of integration testing is to detect any inconsistencies between the software units that are integrated together (called assemblages) or between any of the assemblages and the hardware. System testing is a more limited type of testing; it seeks to detect defects both within the "inter-assemblages" and also within the system as a whole.[16]

5.3 TEST PLAN

A test plan is a detailed document that outlines the test strategy, Testing objectives, resources (manpower, software, hardware) required for testing, test schedule, Test Estimation and test deliverables.

The test plan serves as a blueprint to conduct software testing activities as a defined process which is minutely monitored and controlled by the test manager.[17]

5.3.1 Testing Methods

Below are the main approaches were used to test the developed system in the testing process.

- Black box testing
- White box testing

Black-box testing is a method of software testing that examines the functionality of an application without peering into its internal structures or workings. This method of test can be applied virtually to every level of software testing: unit, integration, system and acceptance. It is sometimes referred to as specification-based testing.[18]

White-box testing (also known as clear box testing, glass box testing, transparent box testing, and structural testing) is a method of testing software that tests internal structures or workings of an application, as opposed to its functionality (i.e. black-box testing). In white-box testing an internal perspective of the system, as well as programming skills, are used to design test cases[19]

5.3.2 Testing Levels

Some testing levels were used to test the developed system.

- Unit Testing
- Integration Testing
- System Testing

Unit Testing

In computer programming, unit testing is a software testing method by which individual units of source code, sets of one or more computer program modules together with associated control data, usage procedures, and operating procedures, are tested to determine whether they are fit for use.[20]

Integration Testing

Integration testing (sometimes called integration and testing, abbreviated I&T) is the phase in software testing in which individual software modules are combined and tested as a group. It occurs after unit testing and before validation testing. Integration testing

takes as its input modules that have been unit tested, groups them in larger aggregates, applies tests defined in an integration test plan to those aggregates, and delivers as its output the integrated system ready for system testing. [21]

System testing

System testing of software or hardware is testing conducted on a complete, integrated system to evaluate the system's compliance with its specified requirements. System testing falls within the scope of black-box testing, and as such, should require no inner design of the code or logic.[22]

5.4 TEST CASES

5.4.1 Login Model

ID	Test Description	Steps to Test	Expected Results	Priority
01	Log-in to the dashboard.	Enter valid username and password and click “Login” button.	Success: Display user Fail: Display an error message	High
02	Login into user	Enter invalid username or password or both and click “Login” button again.	Success: redirect to dashboard. Fail: Display an error message which says “Invalid username or password”.	High
03	Logout from the dashboard.	Click the “logout” button.	Redirect into the login page after the user logout from the dashboard.	High

04	Display logged username and the member information.	Identify the correct username on the header and the information on the page.	Show current username on the header and the User information on the page.	High
----	---	--	---	------

Table 5. 1-Login Model

5.4.2 User Model

01	Create new user.	Administrator or operator should give the username and password and fill the information form for the relevant user.	Administrator & Operator should be able to register new user and show invalid field inputs to refill the form.	High
02	Edit or update member information when required.	After login to the dashboard information could be edited or updated.	Update the user information and show invalid inputs using error messages.	High
	View user	User and their information could be viewed by clicking "View User" on the dashboard menu.	Display User and their relevant information.	Medium
	Invalid controller access.	Enter invalid URL or controller name.	Display an error message	High
	Get back to dashboard	Click "Home" link on the left panel.	Redirect to the site dashboard	High

Table 5. 2-User Model

5.4.3 Form Module

ID	Test Description	Steps to Test	Expected Results	Priority
01	Test data validation by using valid data.	Input valid data and fill each and every fields.	Submit form successfully and show the message “Added Successfully”.	High
02	Test data validation by using invalid data.	Input invalid data and leave blank fields.	Display an error message according to the relevant error that was made.	High
03	Add information to each and every form	Select relevant form and fill each field by adding data then submit it.	Submit information to the database and redirect to the dashboard and show messages for valid or invalid data.	High

Table 5. 3-Form Model

5.4.4 Reports Module

ID	Test Description	Steps to Test	Expected Results	Priority
01	Report Creation	Select relevant field and Click on “Generate Report” button.	The system will display the correct report according to the given data.	High
02	Test report creation using valid data.	Select valid parameters according to the report and Click on “Generate Report” button.	Display valid reports on the screen.	High

03	Test report creation using invalid data.	Select invalid parameters according to the report and Click “Generate Report” button.	Display an error message for invalid parameter selection.	High
----	--	---	---	------

Table 5. 4-Reports Model

5.5 Test Data and Test Results

Test results play a major role on success or the failure of a project. Regular tests were done accordingly throughout the development and the testing stages. Success or the failure of the project can be decided on the test results compared to expected test results.

5.5.1 Acceptance Testing

An acceptance test is a formal description of the behavior of a software product, generally expressed as an example or a usage scenario. A number of different notations and approaches have been proposed for such examples or scenarios. In many cases the aim is that it should be possible to automate the execution of such tests by a software tool, either ad-hoc to the development team or off the shelf.

Similar to a unit test, an acceptance test generally has a binary result, pass or fail. A failure suggests, though does not prove, the presence of a defect in the product.

Teams mature in their practice of agile use acceptance tests as the main form of functional specification and the only formal expression of business requirements. Other teams use acceptance tests as a complement to specification documents containing uses cases or more narrative text.

Teams mature in their practice of agile use acceptance tests as the main form of functional specification and the only formal expression of business requirements. Other teams use acceptance tests as a complement to specification documents containing uses cases or more narrative text.

The Acceptance test were carried out to get the feedback about the developed system from system users. Some of the system users like principal and the warden were participated in the acceptance testing. The users were given questionnaires for each and every module of the system. Below are the questionnaires given to the users and the feedbacks received after the test.[23]

No	Questions	Very good	Good	Satisfied	Poor
01	Design of the system interfaces		Good		
02	Security aspects of the system information.	Very good			
03	Easiness of using the system.	Very good			
04	Proper form validations when filling forms of the system		Good		
05	The ease of generating reports using the system.	Very good			
06	Understandability of buttons, links, icons, menus and popup windows.	Very good			
07	Capability of understanding the error messages.			Satisfied	
08	Overall capability of understanding the system.				Poor
09	The ease of generating reports using the system.				
10	Overall feeling about the system.		Good		

Table 5. 5-Questionnaire for User Evaluation.

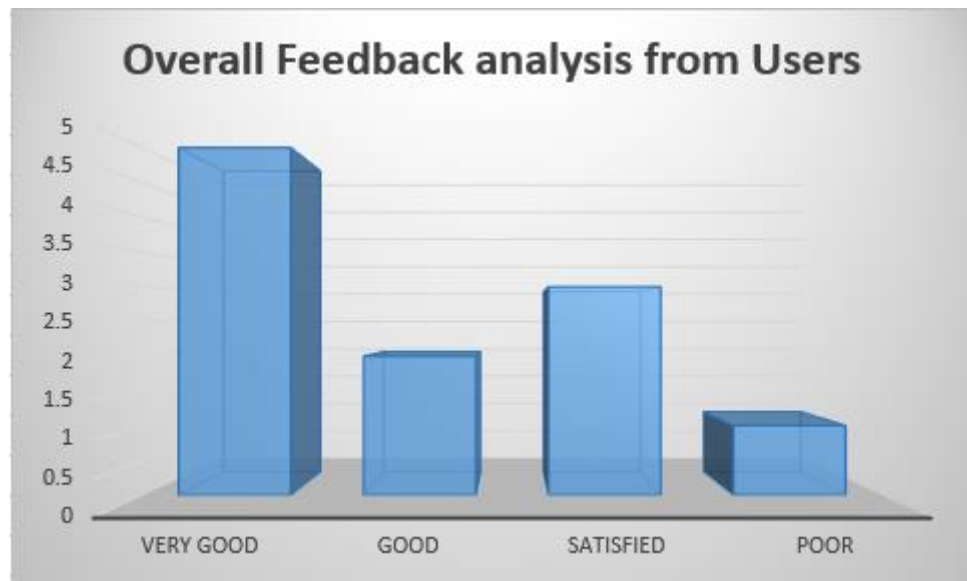


Figure 5. 1-Overall feedback analysis from users

CHAPTER 6 - CONCLUSION

6.1 Chapter Overview

This Chapter is contained in details about conclusion also the critical assessment of the project, problem encounter, lessons learnt and future work regarding the system.

6.2 Overview

Visakha Vidyalaya is one of the leading girl's schools in Sri Lanka. The Hostel also located in school premises. Security and facilities of the Hostel is in an excellent level. They do care about the comfort and needs of the students throughout the whole past years until now.

The hostel is maintained traditional manual system for record student details and hostel details. But the hostel staff, student and also their parents find many difficulties using the system and it was really hard to keep past records of students and relevant data which are affected to them since there are lack of space to keep such amount of hard copies. An online system was much needed for the system because of these difficulties.

Manual records were observed very carefully when gathering requirements to have the exact idea of what the system needs to be added when implementing. Since they have not gone through a web-based system, the functionalities must be given as they expect.

The human mistakes specially occur while handling payments can be avoided using the developed system since the system catches every details and it is generated a report regarding payments. User feedback and some of testing methods were used to avoid conflicts that may occur while using the system by the end users who validate the overall system. User feedback, system functionalities and testing results were used to identify the satisfaction level compared to current manual system.

The system was developed using PHP, according to MVC architecture and other development languages such as HTML, CSS and JS.

6.3 Problems Encountered

The major problems were occurred during the development was the hostel staff and the student were having lack of knowledge regarding a developed system since they have not use it before. Collaborating with the hostel staff and their methods and also the subject knowledge of the supervisor helped to overcome those issues.

6.4 Lessons Learnt

This project was a great opportunity to have the practical knowledge in software development from inception phase to the transition phase. The real customer involvement and the experience gained working in the customer environment was a great opportunity to one can have.

This project was helped to test and implement most important theories and technologies and also the technical knowledge gained while developing the project was priceless. Some technologies were learnt during the BIT degree program and some of them were learnt by observing internet by referring online documentations. Some technologies such as AJAX, JSON and concepts like object oriented were found and learnt to make this development much better. Learning MVC design pattern were given more effort during the process

The dissertation writing was help to gain knowledge about the way of writing a professional document which may help to our future endeavors. I believe the knowledge and the all the experienced I was given during the developing of the project is an outstanding step for my career.

6.5 Future Enhancement

The system will send students' parents or guardians about the monthly payments. The parents will get a message to their mobile phone reminding monthly hostel payments and relevant extra payments if needed. This function is very useful and it may help to avoid some issues regarding the payments. The system will help to make this happen in near future.

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APPENDIX A-

SYSTEM

DOCUMENTATION

This documentation is to help for the system administrators who wish to implement and enhance this system, to install and configure the system in technical perspective. The system documentation can be referred if there are any changes to be made in the Daily Tasks Management System. In order to install the system, the device chosen for installation should meet the following minimum requirements of hardware and software

Hardware Requirements

Hardware	Recommended Minimum Requirement
Processor	1.8GHz Intel core 2 Duo or similar or newer Processor.
Memory	1GB RAM or more
Hard Disk	Minimum 1GB free disk space or Higher.
Printer	Dot-matrix printer or Ink jet printer or Laser Printer.
Internet	Minimum 512kbps ADSL connection.

Table A. 1-Minimum hardware requirements

Software Requirement

Softw are	Recommended Minimum Requirement
Operating system	Microsoft Windows XP/Vista/Windows7 /Windows 10
Bundle Package Software	WampServer 2.4 or Individual packages-MySQL 5.6, PHP 5.4, Apache 2.4.
Image Editor	Adobe Photoshop CS6 or CC
Web browser	Chrome, Firefox, Safari, Internet Explorer or Opera

Table A. 2-Minimum software requirements

Installing WampServer (Apache server, PHP, MySQL)

WAMP server should install first before setup the system in the computer. The steps to install the WAMP server described below.

STEPT 1 - Install WAMP server

- Download WAMP Windows from Download WAMP server from <http://www.wampserver.com/en>



Figure A. 1-WAMP server download link

STEP 2 – Re-store SQL file

- Open the web browser and type <http://localhost/phpmyadmin> then navigate.
- Log in to the database and provide a username (“root”) and password if you want to set a password in the address bar.
- Create new empty database named “db_vvhostel” Click the “Import” button to import files.
- Open the CD and select the “db_vvhostel” file locates inside the Database folder. Then Ccick “Go” button.

STEP 3 - Launching the system

- Open the web browser and type [http://localhost/ db_vvhostel](http://localhost/db_vvhostel) and navigate. Enter valid username and password to log in to the system

APPENDIX B - DESIGN DOCUMENTATION

Use case Diagrams with Use Case Narratives.

Following tables show the use case narratives which were designed for the use case Diagram in chapter 3.

Use case	Log in
Actors	Warden, Vice Principal, Parent, Student, Security
Overview	
Login details are entered by the registered system user and login to the system	
Precondition	
User must have valid user account	
Event Flow	
1. User enter login credentials 1.1 Enter invalid login credentials and system displays an error message 2. The system users are authenticated 3. User has forgotten the password and reset the password 4. User enter valid credentials and check authentication and log in and direct to user dashboard	
Post Conditions	
User can access system privileged data.	

Table B. 1-Use case narrative for user log in

Use case	Registration of Student
Actors	Student
Overview	
The registration form is provided to all students in the school and it will be checking by the system and all registered students are identified as new registrations	
Precondition	
Student must be currently studying in the school as a permanent student.	
Event Flow	
1. Student must fill the registration form via online by providing necessary information 2. System verifies that all information is in the correct format. 3. Student will receive a success message with an email.	
Post Conditions	
Student will be added to the system as a new registration.	

Table B. 2-Use case narrative for Student Registration

Use case	Make Visitor Appointment
Actors	Visitor
Overview	
Relevant appointment details are entered by the system user in order to make an appointment	
Precondition	
The visitor and their details must include in the relevant student account.	
Event Flow	
1. Visitor must fill the Appointment form via online by providing necessary information 2. System verifies that all information is in the correct format. 3. Visitor will receive a success message with a confirmation email including visiting date and time.	

Post Conditions
Visitor should confirm the details and participation

Table B. 3-Use case narrative for Make Visitor appointments

Use case	Check-out from the Hostel
Actors	Visitor, Student, Security
Overview	
Student fill the check-out and submit in order to check-out from the Hostel	
Precondition	
A visitor who's taking student out of the Hostel must be in the student account.	
Event Flow	
1. Fill the check-out form using valid details 2. The system verifies details and grant permission to go out from the hostel premises 3. Security checks the details before granting permission to go out from the school premises	
Post Conditions	
Security mark the student as check-out from School.	

Table B. 4-Use case narrative for Check-out from Hostel

APPENDIX C - USER MANUAL

This documentation provides the overall idea of critical and main functionalities of the developed system to the administrator. The system is developed for the internal and external environment of the school.

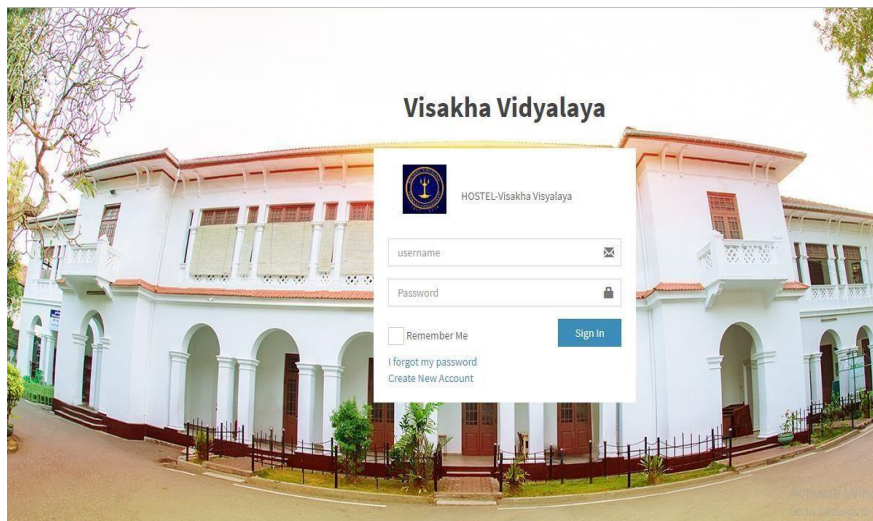


Figure C. 1--Log-in Interface

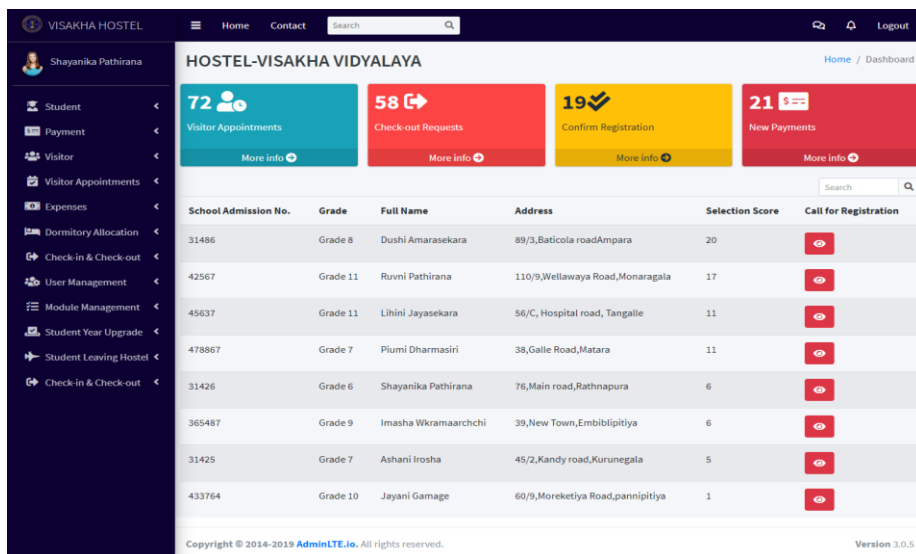


Figure C. 2-Admin Dashboard

[Home](#)
[Contact](#)

Shayanika Pathirana

[Student](#)
[Payment](#)
[Visitor](#)
[Visitor Appointments](#)
[Expenses](#)
[Dormitory Allocation](#)
[Check-in & Check-out](#)
[User Management](#)
[Module Management](#)
[Student Year Upgrade](#)
[Student Leaving Hostel](#)
[Check-in & Check-out](#)

HOSTEL-VISAKHA VIDYALAYA

[Home](#) / [Dashboard](#)

Register Student

Personal Information

Full Name

Piumi Dharmasiri

Calling Name

Piumi

Date of Birth

05/06/2002

Address

38,Galle Road,Matara

District

Matara

Divisional Secretariat

Weligama

School Admission Number

478867

Grade

Grade 7

House

Motwani

Parent/Guardian Information

Name

Anura Dharmasiri

NIC Number

596654567V

Address

38,Galle Road,Matara

Contact Number

0776544896

Occupation

Bank Manger

Email Address

shayanika.vv@gmail.com

In-case of Emergency

Name

Sarath Dharmasiri

NIC Number

647876563V

Address

23/B,University Road, Wijerama

Contact number

07143456789

Occupation

Teacher

Relationship with Student

Uncle

Submit

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Figure C. 3- Registration form

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VISAKHA HOSTEL

Home Contact

Search

Logout

Shayanika Pathirana

Student

Payment

Visitor

Visitor Appointments

Expenses

Dormitory Allocation

Check-in & Check-out

User Management

Module Management

Student Year Upgrade

Student Leaving Hostel

Check-in & Check-out

HOSTEL-VISAKHA VIDYALAYA

Home / Dashboard

Search

In Hostel Premises

Out for School

Check-out from Hostel

	Name	Grade	DOB	Address	Contact Number	Student Info.	Check-out
1	Dushi	grd8	2008-02-08	89/3,Baticola roadAmpara	0712498778		
1	Ashani	grd7	2009-02-08	45/2,Kandy road,Kurunegala	0712498778		
1	Piumi	grd7	2002-05-06	38,Galle Road,Matara	0776544896		
1	Ruvini	grd11	2015-04-13	110/9,Wellawaya Road,Monaragala	0723554673		
1	Jayani	grd10	2004-08-01	60/9,Moreketiya Road,pannipitiya	0723554334		

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Figure C. 4-All student list page

VISAKHA HOSTEL

Home Contact

Search

Logout

Shayanika Pathirana

Student

Payment

Visitor

Visitor Appointments

Expenses

Dormitory Allocation

Check-in & Check-out

User Management

Module Management

Student Year Upgrade

Student Leaving Hostel

Check-in & Check-out

HOSTEL-VISAKHA VIDYALAYA

Home / Dashboard

Search

Confirmed Registration

Rejected Registration

	Student Name	Grade	Register Date	Register Time		
-1	Imasha Wkramaarchchi	Grade 9	2021-01-29	10:22:00		
2	Dushi Amarasekara	Grade 8	2021-01-20	10:26:00		
2	Ashani Irosha	Grade 7	2021-01-20	13:30:00		
2	Piumi Dharmasiri	Grade 7	2021-01-29	23:21:00		
-1	Shayanika Pathirana	Grade 6	2021-01-20	08:30:00		
2	Ruvni Pathirana	Grade 11	2021-01-29	12:21:00		
2	Jayani Gamage	Grade 10	2021-01-29	09:30:00		

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Figure C. 5-Admission called student list

VISAKHA HOSTEL

[Home](#)
[Contact](#)

[Logout](#)

Shayanika Pathirana

[Student](#)
[Payment](#)
[Visitor](#)
[Visitor Appointments](#)
[Expenses](#)
[Dormitory Allocation](#)
[Check-in & Check-out](#)
[User Management](#)
[Module Management](#)
[Student Year Upgrade](#)
[Student Leaving Hostel](#)
[Check-in & Check-out](#)

HOSTEL-VISAKHA VIDYALAYA

[Home](#) / [Dashboard](#)

Student Details

School Admission Number
478867

Grade
Grade 7

Full Name
Piumi Dharmasiri

Date of Birth
2002-05-06

Address
38,Galle Road,Matara

Select District
Matara

Divisional Secretariat
Welligama

Contact Number
0776544896

Email Address
shayanika.vv@gmail.com

Call for Interview form

Registration Date

Registration Time

Documents need to bring

- ☒ School Admission
- ☒ Birth Certificate
- ☐ National identity card / Postal card
- ☒ Postal size photo

Facilities need to bring

- ☐ Laundry basket
- ☒ Carpet
- ☒ White coverlet
- ☒ Mug
- ☒ Mosquito Net
- ☒ Plate
- ☒ Slippers

Email address

[Save & Send Email](#)

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Figure C. 6-Call for Student Admission form


VISAKHA HOSTEL

[Home](#)
[Contact](#)



[Logout](#)


Shayanika Pathirana

[Student](#)
[Payment](#)
[Visitor](#)
[Visitor Appointments](#)
[Expenses](#)
[Dormitory Allocation](#)
[Check-in & Check-out](#)
[User Management](#)
[Module Management](#)
[Student Year Upgrade](#)
[Student Leaving Hostel](#)
[Check-in & Check-out](#)

HOSTEL-VISAKHA VIDYALAYA

[Home](#) / [Dashboard](#)

Visitor Added Successfully..!

Visitor Information

Visitor Name

NIC Number

Address

Contact number

Email Address

Relationship with student

Name	NIC	Address	Contact number	Email	Relationship with student	
Jayantha Chandrasiri	609876676V	60/9,Moreketiya Road,pannipitiya	0723554334	shayanika.vv@gmail.com	Uncle	

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Version 3.0.5

Figure C. 7-Add Visitor Form


VISAKHA HOSTEL

[Home](#)
[Contact](#)



[Logout](#)


Shayanika Pathirana

[Student](#)
[Payment](#)
[Visitor](#)
[Visitor Appointments](#)
[Expenses](#)
[Dormitory Allocation](#)
[Check-in & Check-out](#)
[User Management](#)
[Module Management](#)
[Student Year Upgrade](#)
[Student Leaving Hostel](#)
[Check-in & Check-out](#)

HOSTEL-VISAKHA VIDYALAYA
[Home](#) / [Dashboard](#)

Student Details

Personal Information



Full Name

Dushi Amarasekara

Calling Name

Dushi

Date of Birth

2008-02-08

Address

89/3,Baticola roadAmpara

District

Ampara

Divisional Secretariat

Kalmunai

School Admission Number

31486

Grade

Grade 8

House

Jayathilaka

Parent/Guardian Information

Name

Dilip Amaasekara

NIC Number

614356735V

Address

Rathnapura

Contact Number

0712498778

Occupation

Business

Email Address

shayanika.vv@gmail.com

In-case of Emergency

Name

Swarna Dissanayaka

NIC Number

776789765V

Address

Maharagama

Contact number

6543456789

Occupation

Teacher

Relationship with Student

Aunty

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Figure C. 8-Student Information view

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APPENDIX D - MANAGEMENT REPORT

Below Chart represent the No of Students weekly check-out from Hostel

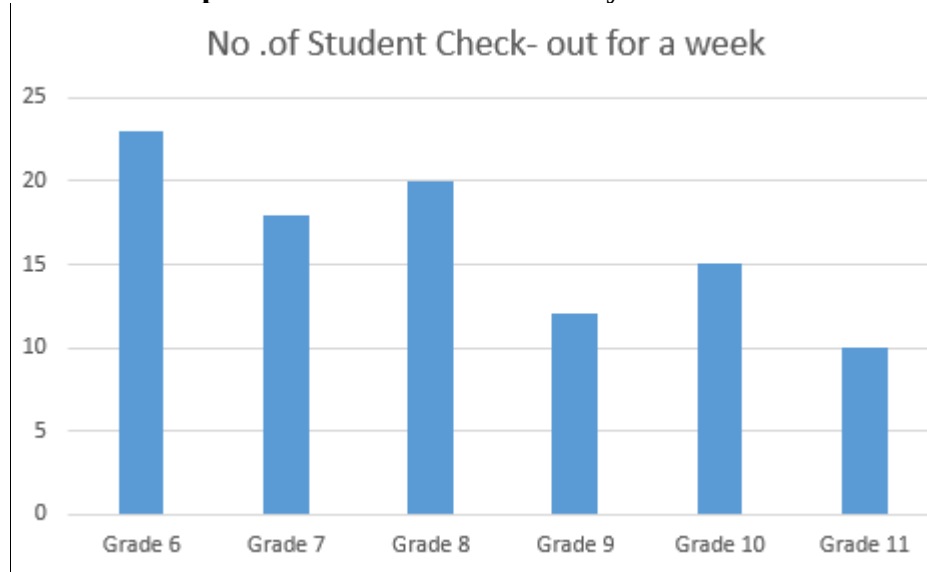


Figure D. 1-No of students check-out for a week

Below Pie chart represent the monthly New admission for the Hostel

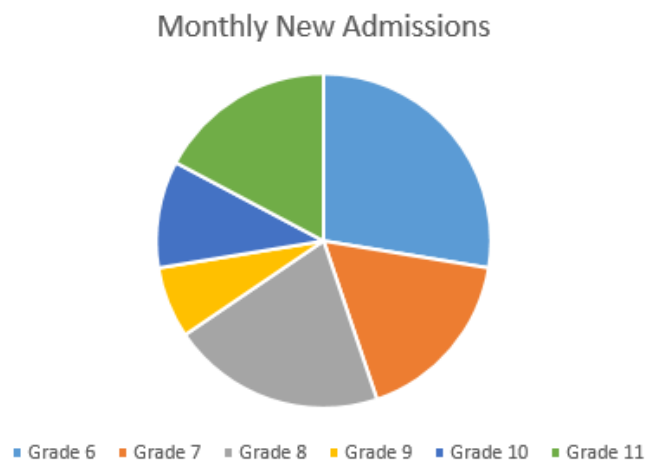


Figure D. 2-Monthly New Admissions

The system can generate below reports

 VISAKHA VIDYALAYA 133, Vajira Road, Colombo - 05				
New Registration Report				
School Admission No.	Grade	Full Name	Address	Selection Score
31486	Grade 8	Dushi Amarasekara	89/3,Baticola roadAmpara	20
42567	Grade 11	Ruvni Pathirana	110/9,Wellawaya Road,Monaragala	17
45637	Grade 11	Lihini Jayasekara	56/C, Hospital road, Tangalle	11
78867	Grade 7	Piumi Dharmasiri	38,Galle Road,Matara	11
31426	Grade 6	Shayanika Pathirana	76,Main road,Rathnapura	6
35487	Grade 9	Imasha Wkramaarchchi	39,New Town,Embiblipitiya	6
31425	Grade 7	Ashani Irosha	45/2,Kandy road,Kurunegala	5
43374	Grade 10	Jayani Gamage	60/9,Moreketiya Road,pannipitiya	1

Figure D. 3-Monthly New Admissions Report

 VISAKHA VIDYALAYA 133, Vajira Road, Colombo - 05			
New Admission Call Report			
Student Name	Grade	Register Date	Register Time
Imasha Wkramaarchchi	Grade 9	1/29/2021	10:22:00
Dushi Amarasekara	Grade 8	1/20/2021	10:26:00
Ashani Irosha	Grade 7	1/20/2021	13:30:00
Piumi Dharmasiri	Grade 7	1/29/2021	11:21:00
Shayanika Pathirana	Grade 6	1/20/2021	8:30:00
Ruvni Pathirana	Grade 11	1/29/2021	12:21:00
Jayani Gamage	Grade 10	1/29/2021	9:30:00

Figure D. 4-New Admission call Report

	VISAKHA VIDYALAYA 133, Vajira Road, Colombo - 05
Payment Receipt	
Date	- 05 / 11 / 2020
Receipt no	- 001104
PAYMENT DETAILS	
• Student id	- G6/102
• Student name	- CHATHURIKA SAGARA
• Month of the payment	- NOVEMBER
• Amount	- 3500/=
• Name of the person	- MRS. DISSANAYAKA
• Name of the warden	- MRS. SUMANA DAMAYANTHI

Figure D. 5-Payment Receipt

APPENDIX E - TEST RESULTS

Login Module

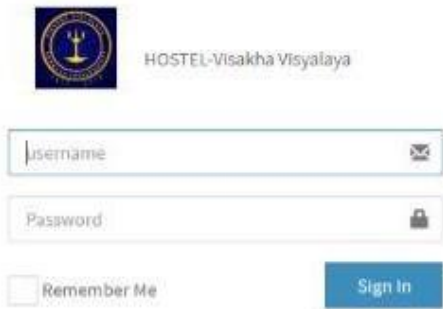
Test No	Test Result	Status
1		Pass
2		Pass
3		Pass

Table E. 1-Log-in module test results

Student Registration

Test No	Test Result	Status
1	Submit without enter Full Name Full Name Please Enter Full Name The Full name should not be empty..!	Pass
2	Enter already exist Admission number School Admission Number 31425 Sorry,this Admission Number already exist	Pass
3	Enter Future date as Date of birth Date of Birth 02/05/2021 Future date cannot use as a date of birth....!	Pass
4	Enter invalid Mobile number Contact Number 07745337860 The Contact Number is Invalid..!	Pass
5	Enter without Email address Email Address Please Enter Email Address The Email Address should not be empty..!	Pass
6		Pass

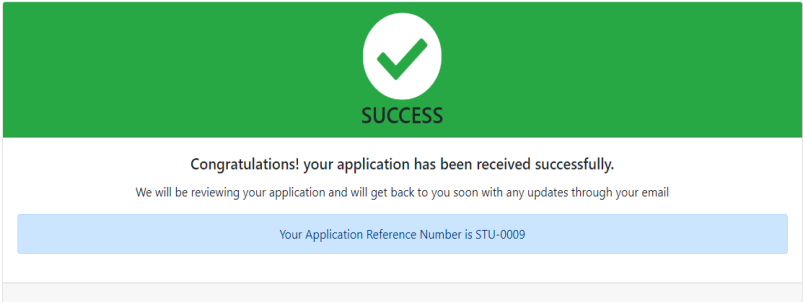
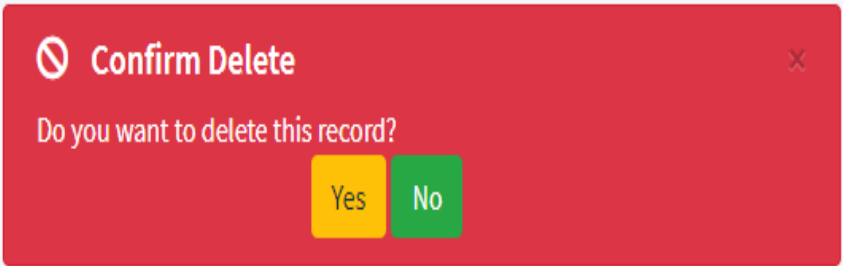
	Register success message 	
7	Upload existing profile image Upload Profile Image Choose File No file chosen Sorry, file already exists..!	Pass
8	Submit Without selecting Grade Grade -Please Select Grade- Please select the Grade..!	Pass
9	Select date on Calendar 	Pass
10	Confirm deleting Record 	Pass

Table E. 2-Student registration module test results

APPENDIX F - CLIENT CERTIFICATE



Figure F. 1-Client Certificate

GLOSSARY

AJAX (Asynchronous JavaScript and XML) - is a group of interrelated web development methods used on the client-side to create interactive web applications. It is the art of exchanging data with a server, and update parts of a web page without reloading the whole page

CSS (Cascading Style Sheet) - is a style sheet language used to describe the presentation semantics (the look and formatting) of a document written in a mark-up language.

Design pattern - is a way of reusing abstract knowledge about a problem and its solution. It is simply a description of the problem and the essence of its solution.

Domain - is knowledge about the environment in which the target system operates. Graphical User Interface - is a type of user interface that allows users to interact with electronic devices with images rather than text commands.

Internet - is a global system of interconnected computer networks that use the standard Internet Protocol Suite (TCP/IP) to serve billions of users worldwide.

Iterative and Incremental Development - is a cyclic approach to software development in which activities are repeated in a structured manner and the software is developed in increments.

XHTML (Extensible Hypertext Mark-up Language) - is the predominant mark- up language for web pages. HTML is the basic building-blocks of web pages, which has collaborated with XML conventions.

JavaScript - is a prototype-based, object-oriented client-side scripting language. It is also considered as a functional programming language.

Object Oriented Development - is a standard approach to software development based on objects and its instances.

PHP Hypertext Pre-processor (PHP) - is a server-side programming language.

HTTPS (Hypertext Transfer Protocol Secure) - is a combination of the HTTP with SSL/TLS protocol to provide encrypted and secured communication.

Unified Modelling Language (UML) - is a standardized general-purpose modelling language in the field of object-oriented engineering. This includes a set of graphic notation techniques to create visual models of object-oriented systems.

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